

Solving Word Problems Involving Volume of Cylinders

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Recall

The volume of a cylinder:

$$V = \pi r^2 h$$

If the diameter is given:

$$r = \frac{d}{2}$$

Use $\pi \approx 3.14$ unless another value of π is specified.

Make sure all measurements are expressed in the same **unit of measure**.

Unit Conversions

Convert first, then compute.

$$1 \text{ m} = 100 \text{ cm}$$

$$1 \text{ cm} = 10 \text{ mm}$$

For example:

$$2 \text{ m} = 200 \text{ cm}$$

Key Words

These phrases ask for the **volume** of a cylinder:

“How much can it hold?”

“Find its capacity.”

“How many cubic meters...?”

Steps in Solving Word Problems

Question: Determine what is asked.

Given: Identify the given values.

Formula: Choose the appropriate formula.

Compute: Perform the calculations.

Answer: Write the answer with the correct unit.

Memory Tip

**Quick Green Frogs Catch
Ants**

Question

Given

Formula

Compute

Answer

Examples

Example 1

A cylindrical water tank has a radius of 4 m and a height of 5 m. Find its volume.

$$\begin{aligned} V &= \pi r^2 h \\ &\approx 3.14(4)^2(5) \\ &\approx 3.14(80) \\ &\approx 251.2 \text{ m}^3 \end{aligned}$$

Example 2

A cylindrical can has a diameter of 10 cm and a height of 12 cm. How much can it hold?

$$r = \frac{10}{2} = 5 \text{ cm}$$

$$\begin{aligned} V &= \pi r^2 h \\ &\approx 3.14(5)^2(12) \\ &\approx 3.14(300) \\ &\approx 942 \text{ cm}^3 \end{aligned}$$

Example 3

A pipe has an inside radius of 3 cm and a length of 2 m. Assuming it is a cylinder, determine its volume.

$$2 \text{ m} = 200 \text{ cm}$$

$$\begin{aligned} V &= \pi r^2 h \\ &\approx 3.14(3)^2(200) \\ &\approx 3.14(1\,800) \\ &\approx 5\,652 \text{ cm}^3 \end{aligned}$$

Example 4

A cylindrical drum can hold 628 cm^3 of oil. If its radius is 2 cm, find its height.

$$V = \pi r^2 h$$

$$628 \approx 3.14(2)^2 h$$

$$628 \approx 12.56h$$

$$h \approx 50 \text{ cm}$$

Example 5

A cylindrical container has a volume of 2 200 cm³. If its height is 7 cm, find its radius. Use $\pi = \frac{22}{7}$.

$$V = \pi r^2 h$$

$$2\,200 = \frac{22}{7} r^2 (7)$$

$$2\,200 = 22 r^2$$

$$r^2 = 100$$

$$r = 10 \text{ cm}$$

Example 6

A cylindrical jar has a radius of 14 cm and a height of 25 cm. Water costs ₱10 per 1 000 cm^3 . How much will it cost to completely fill the jar? Use $\pi = \frac{22}{7}$.

Example 6: Volume

$$V = \pi r^2 h$$

$$= \frac{22}{7} (14)^2 (25)$$

$$= 15\,400 \text{ cm}^3$$

Example 6: Cost

Number of 1 000 cm³ units:

$$15\,400 \div 1\,000 = 15.4$$

Cost:

$$15.4 \times 10 = 154$$

The cost is ₱154.

Practice Exercises

Practice 1

A cylindrical water tank has a radius of 5 m and a height of 8 m. Find its volume.

Practice 2

A cylindrical container has a diameter of 14 cm and a height of 20 cm. Determine its volume.

Practice 3

A silo has a radius of 6 m and a height of 15 m.
How many cubic meters of grain can it hold?

Practice 4

A cylindrical flower pot has a diameter of 24 cm and a height of 30 cm. Find its capacity.

Practice 5

A cylindrical drum can hold 1570 cm^3 of oil. If its radius is 5 cm, find its height.

Practice 6

A cylindrical container has a volume of 2 464 cm^3 . If its height is 16 cm, find its radius. Use

$$\pi = \frac{22}{7}.$$

Activities

Activity 1

A juice can has a radius of 3 cm and a height of 15 cm. How much juice can it hold?

Activity 2

A pillar is shaped like a cylinder with radius 4 cm and height 25 cm. Find its volume.

Activity 3

A storage silo has a radius of 8 m and a height of 12 m. How many cubic meters of corn can it hold?

Activity 4

A cylindrical bucket has a diameter of 28 cm and a height of 35 cm. Find its capacity.

Activity 5

A pipe has an inside radius of 3 cm and a length of 90 cm. Assuming it is a cylinder, determine its volume.

Activity 6

A cylindrical drum can hold $2\,772\text{ cm}^3$ of liquid. If its radius is 7 cm , find its height. Use $\pi = \frac{22}{7}$.

Activity 7

A cylindrical aquarium has a radius of 28 cm and a height of 75 cm. Water costs ₱18 per 1 000 cm³. How much will it cost to completely fill the aquarium? Use $\pi = \frac{22}{7}$.

Thank you!